Quiz #2 (NLP)

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Q1: Given 3-document corpus is

d1 : information retrieval is process of index search retrieval
d2 : retrieval is used for evaluation of search results retrieval retrieval
d3 : evaluation in information in evaluation process search

(a) Convert d2 in vector form using raw counts as weights. Clearly mention dimensions and its value.

(b) Convert the words 'results' in form of vector. Context window size is 2. Use TF-IDF weights.

Q2: Given confusion matrix is

System output	Gold labels: Coffee	Tea	Beans
Coffee	10	30	15
Tea	40	5	25
Beans	20	40	300

1) Microaverage (Precision, Recall, F-Score)
2) Macroaverage (" " ")

Q1(a) For d2 we get

< information : 0
retrieval : 3
is : 1
process : 0
of : 1
index : 0
search : 1
used : 1
for : 1
evaluation : 1
results : 1
in : 0 >

Q1 (b) For word results

words	information	retrieval	is	process	of	index	search	used	for	evaluation	results	in
Raw Counts	0	2	0	0	1	0	1	0	0	0	1	0
TF	0	1.301	0	0	1	0	1	0	0	0	1	0
IDF	0.176	0.176	0.176	0.176	0.176	0.477	0	0.477	0.477	0.176	0.477	0.[illegible]
TF×IDF	0	0.2289	0	0	0.176	0	0	0	0	0	0.477	0

where $TF = 1 + \log(tf)$ $\quad IDF = \log\left(\frac{N}{df}\right)$

Q2

	Coffee	Tea	Beans
Coffee	10	30	15
Tea	40	5	25
Beans	20	40	300

Coffee

10	45
60	370

$$\text{Precision} = \frac{10}{10+45} = \boxed{0.182}$$

$$\text{Recall} = \frac{10}{10+60} = \boxed{0.143}$$

$$\text{F-Score} = \frac{2(0.182)(0.143)}{0.182+0.143} = \boxed{0.160}$$

Tea

5	65
70	345

$$\text{Precision} = \frac{5}{5+65} = \boxed{0.071}$$

$$\text{Recall} = \frac{5}{5+70} = \boxed{0.067}$$

$$F_{tea} = \frac{2(0.067)(0.071)}{0.067+0.071} = \boxed{0.069}$$

Beans

300	60
40	85

$$\text{Precision} = \frac{300}{300+60} = \boxed{0.833}$$

$$\text{Recall} = \frac{300}{300+40} = \boxed{0.882}$$

$$F_{beans} = \frac{2(0.833)(0.882)}{0.833+0.882} = \boxed{0.856}$$

Pooled

315	170
170	800

$$\text{Precision} = \frac{315}{315+70} = 0.818$$

$$\text{Recall} = \frac{315}{315+170} = 0.649$$

$$\text{F-Score} = \frac{0.818 \times 0.649}{0.818+0.649} = \frac{2(0.531)}{1.467} = 2(0.362) = 0.7239$$

Macro average Results

$$\text{Precision} = \frac{0.182+0.071+0.833}{3} = \boxed{0.362}$$

$$\text{Recall} = \frac{0.143+0.067+0.882}{3} = \boxed{0.364}$$

$$\text{F-Score} = \frac{0.16+0.069+0.856}{3} = \boxed{0.361}$$

Micro average Results

Precision = 0.818

Recall = 0.649

F-Score = 0.724